Does Remote Sensing Beat the Trail Map? Predicting Hiking Speed in Unseen Regions from Lidar Terrain and OpenStreetMap Tags

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***Abstract*—[TODO: 150–200 words, written last. Problem: travel time on foot; Tobler. Data: 17 regions, 290k walking segments from public OSM GPS traces. Test: leave one region out, calibrated predictive distributions (CRPS). Result: terrain from airborne-lidar elevation transfers to unseen regions (+2.1 to +2.8% CRPS skill over a refitted slope curve), map tags barely do (+0.4 to +0.9%), satellite embeddings help within regions but transfer less reliably; Tobler is 12.5% worse. A terrain-adjusted hiking function needs one DEM-derived input.]**

***Index Terms*—hiking speed, lidar, OpenStreetMap, GPS trajectories, transfer, calibration**

I. Introduction

[TODO: Draft by Claude: rewrite in your own words.]

How long a walk will take matters for search and rescue, for wildland firefighters' escape routes [1] and for the times trail apps show. A common starting point is Tobler's hiking function [2], $v = 6\exp(-3.5\,|s + 0.05|)$ km/h on slope s, which peaks at 6 km/h on a gentle descent. It is still in use, from maps of what a person can reach on foot [3] to traverse planning for planetary exploration [4].

Crowdsourced GPS tracks show that the function is too fast and too peaked, and that speeds at one slope vary widely [5], [6]. Newer models add covariates: OSM terrain classes and lidar-measured obstruction in England and Wales [7]; boosted trees on 16 factors, weather included, in Hong Kong [8]; a latent walking speed predicted from Sentinel-2 imagery in one Croatian range [9]; the hiker's pace on the first part of a hike [10]. STRIDE [11] models walking speed from lidar slope, understory density [12] and ground roughness, calibrated on timed field walks. On roads, speeds and their spread have been learned from overhead imagery and lidar [13], [14].

Wood et al. [7] test with 10-fold cross-validation and Huang et al. [8] with a random split, so their test tracks come from the areas they trained on. Yet predictions matter most where nobody has recorded a hike: in the White Mountains, one of our best-sampled regions, 77% of the 3,369 km of mapped trail has no trace in our data. Two sources could describe such trails. On the trail map, OSM mappers tag difficulty (sac_scale), surface and visibility, but unevenly: sac_scale is on 1–96% of walked trail, depending on the region. Remote sensing reaches every trail, tagged or not: the USGS 3D Elevation Program collects airborne lidar across the United States [15], and satellite embeddings summarize imagery worldwide [16]. To our knowledge, no study has tested which of the two works in regions a model has never seen.

We test it on public OpenStreetMap GPS traces [17] from 17 mountain regions of the United States and Canada. We contribute (i) a dataset of 290,640 walking traversals of 50 m trail segments (1,992 outings) with slope, OSM tags and remote-sensing covariates (Sec. II); (ii) a transfer test that holds out whole regions, zones or half the continent and scores predictive distributions with the continuous ranked probability score (CRPS) [18], against published hiking functions (Sec. III); and (iii) the finding that terrain measured from elevation data transfers ($+2.1$ to $+2.8\%$ CRPS skill over a refitted slope curve in unseen regions, ahead of tags in 16 of 17), while tags add $+0.4$ to $+0.9\%$ and neither vegetation, from 30 m maps or lidar point clouds, nor satellite embeddings match it (Sec. IV). Tobler's function scores worse than the refitted curve in every region (-12.5% overall); for the same hiker on the same trail, descending is 8–23% faster than climbing, not the 42% Tobler implies; and over whole hikes the hiker's own pace matters more than any covariate. We also give a Tobler-style function with one DEM-derived input ($+1.0\%$).

II. Data

[TODO: Draft by Claude: rewrite in your own words.]

Sources. We use public GPS traces uploaded to OpenStreetMap (OSM) [17] in 17 mountain regions of the United States and Canada (Fig. 1), chosen for relief, trail density and lidar coverage. The traces come from three sources: the OSM trackpoints API, which returns the timestamped points of identifiable and trackable uploads tile by tile; the original GPX files of identifiable uploads, which give whole hikes at the device's own sampling; and a crawl of 1.5 million original files of public and identifiable uploads from 2016–2022, which keeps the timestamps the API drops. Sixteen regions combine all three; the Bay Area uses the crawl only, because the API times out on its tiles.

Processing. Each recording is cleaned (repeated timestamps, steps faster than 60 km/h), split at gaps over 5 min and thinned to points at least 5 s apart. A step counts as a stop when the point 60 s later lies within 10 m. Traces are map-matched to the OSM trail and road network with a hidden Markov model [19] (97% of points matched), and the matched path is cut into 50 m segments. Each segment records its moving speed, its signed slope along the direction of travel from the 5 m USGS 3DEP elevation grid [15] (resampled from 1 m lidar

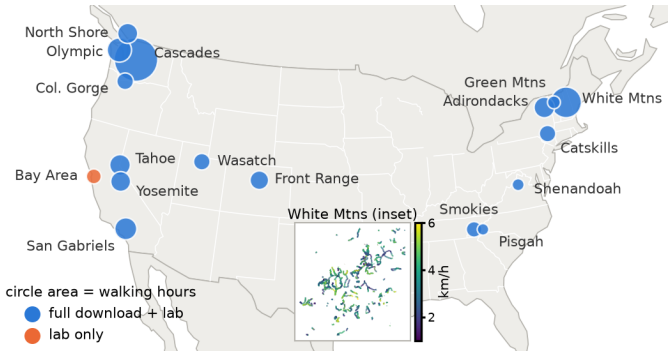


Fig. 1. The 17 regions. Circle area is walking hours (1,992 outings, 4,702 h in total). Blue: full OSM download plus the lab crawl; orange: lab crawl only. Inset: the White Mountains' 34,535 walking segments colored by speed.

under at least 83% of walked trail in 13 regions), the OSM tags of its way and the trace's recording interval. Duplicate uploads are removed, and traces that start within 15 min and 1 km of each other form one outing (a group hike), the unit we resample. Travel mode is inferred per trace, anchored on trails closed to bikes and horses. Against the activity that 327 uploaders' apps wrote into their files, 93% of labelled hikes are classed as walking, and 5% of the labelled traces we class as walking say bike or ski.

The dataset. The regions hold 2,876 hiking traces. The walking analysis set (walking traces on trails) has 290,640 segments (14,532 km) from 2,046 traces in 1,992 outings and 4,702 moving hours. Its uploaders are OSM mappers and enthusiasts rather than a random sample of hikers, and busy API tiles do not reach their oldest uploads. Traces logged more than 10 s apart read up to 11% slow, because straight lines between points cut corners and short stops hide between them; the gains we report are the same at every recording rate (Sec. IV).

III. Methods

[TODO: Draft by Claude: rewrite in your own words.]

Target and covariates. We predict the distribution of $\ln v$, the log moving speed of a walking segment. Covariates come in five groups: (a) slope; (b) OSM tags: sac_scale, trail_visibility, surface, width, smoothness, mtb_scale, highway type and permitted modes; (c) terrain from elevation, read every 1 m along the segment's matched path: the spread (SD) and maximum of its 5 m pitches, the relief around an 11 m moving average and the side slope, from the 5 m grid and, where available, 1 m lidar, which adds STRIDE's ground roughness [11]; (d) 2021 land cover and tree canopy at 30 m [20]; (e) 2022 AlphaEarth satellite embeddings at the segment midpoint [16], [21]. As a control, "hidden climbing" is the extra time Tobler's function predicts when the segment is walked as ten 5 m pieces instead of at its average grade.

Models. The baselines are Tobler's function [2], the published median curve of Campbell et al. [6], STRIDE [11] as published and with its form refitted, and our reference: a rounded, asymmetric slope curve refitted to the training data, $\ln v = a - k_{\pm}\left(\sqrt{(s+s_0)^2+e^2}-e\right)$, with k_+ above the peak and k_- below it. Each covariate group then enters two models: an additive one (ridge regression of the curve's residuals on one-hot bins: deciles of continuous covariates, one level per tag value plus "missing") and gradient-boosted trees on slope and the covariates. The terrain-adjusted hiking function keeps Tobler's form but lets peak speed and slope sensitivity depend on one DEM-derived number, the steppiness z:

$$\ln v = a_0 + a_1 z - (k_0 + k_1 z)\,|s + s_0|, \qquad (1)$$

with z the SD of the 5 m pitches along the path divided by 0.10.

TABLE I
CRPS skill (%) vs the refitted slope curve. Additive / trees; 95% intervals for unseen regions resample whole regions.

Model	New hikers	New 10 km	**Unseen region**
Tobler 1993	−12.8	−12.7	−12.5
Campbell 2022	−0.9	−0.9	−0.5
STRIDE 2024 (refit)	−17.2 (+0.2)	−17.3 (+0.2)	−16.8 (+0.2)
OSM tags	+1.3 / +1.5	+1.1 / +1.5	+0.9 / +0.4
30 m vegetation	+0.0 / +0.1	−0.0 / −0.1	−0.1 / −0.8
Embeddings	+1.4 / +2.6	+1.4 / +1.9	+1.1 / +1.1
Lidar vegetation†	+0.4 / +0.5	+0.4 / +0.4	+0.3 / +0.2
Terrain	+2.2 / +3.2	+2.1 / +2.8	**+2.1 / +2.8**
Terrain + emb. + tags	+3.7 / +5.1	+3.8 / +5.1	+3.0 / +3.9
Eq. (1), one input	+1.0	+1.0	+1.0

†The 15 regions with lidar point clouds.

Predictive distributions and tests. Every model predicts its point estimate plus the quantiles of its residuals on held-out outings of the training data (training outings split 70/15/15 into fitting, early stopping and calibration), so calibration has to transfer too. Splits never separate a segment from its hike: 5 folds of outings (new hikers), 5 folds of 10 km UTM blocks (new places) and, the headline, leave one region out (17 folds); stricter checks hold out whole zones or half the continent. We report the continuous ranked probability score (CRPS) [18] of $\ln v$ as skill against the refitted slope curve, $1 - \mathrm{CRPS}/\mathrm{CRPS}_{\mathrm{slope}}$, with 95% intervals that resample outings (whole regions for unseen regions); the coverage of 90% intervals; and the median error in moving time over whole hikes after their first kilometre. A reference shows how much is the hiker: the median residual of a hike's first kilometre, shrunk by a factor fitted on calibration outings (0.61), applied to the rest of the hike. Asymmetry is measured on out-and-back pairs, the same outing on the same OSM way in both directions (mean grade at least 3%), as the ratio of speed down to speed up.

IV. Results

[TODO: R1 (Fig. 2a, Table I): Tobler fails everywhere (−12.5%). Walking speed peaks near 4.2 km/h on level to gently descending trail, not 6: Tobler is ∼20% too fast within ±10% grade and ∼18% too slow on climbs steeper than 25%. Campbell's published median curve matches near level but is ∼20% too fast on climbs steeper than 25%; it ties our refit on

average, ranging by region from −3.8% (White Mountains) to +5.8% (Shenandoah).]

[TODO: R1b, asymmetry (Fig. 2b): on 2,763 out-and-back pairs (same hiker, same trail; 1,145 outings) descending is 8% faster than climbing at 3–10% grade, 22–23% above 20%; Tobler implies 42% at every grade above 5%. Our pooled curve implies nearly the same ratios as the pairs, so its asymmetry is not a sampling artifact; Campbell's curve implies less (2–16%).]

[TODO: R2: tags do not transfer: +1.3% within regions, +0.9% / +0.4% in unseen ones.]

[TODO: R3: terrain transfers: +2.1% / +2.8% in unseen regions, beating tags in 16 of 17 (Fig. 3b); terrain minus tags +0.7 to +2.2, above zero in every bootstrap draw. About 60% of the gain is uneven grade inside the segment. Not a region proxy (adds ∼1.9 points over region offsets), not a hiker proxy. 1 m lidar ≈ 5 m; STRIDE's roughness adds nothing on trails; 30 m vegetation adds nothing; embeddings help within regions but transfer less reliably. Understory density and canopy height from the lidar point clouds (15 regions) help within a region (White Mountains: +3.6% → +5.0% in new 10 km squares) but not in unseen regions (+0.2 points over terrain, helping in 9 of 15); STRIDE's published density term over-penalises trails (−18% → −35%).]

[TODO: R3 robustness: the gain holds for newer hikes (trained up to 2019, tested on 2020 and later: +2.2% / +2.9%; tags +0.9%) and at every GPS recording rate (unseen regions, traces every ≤2 s, 2–10 s, >10 s: +1.7, +2.4, +1.9%). By slope it is largest on descents and level trail (+2.3 to +2.8%) and smallest on steep climbs (+0.8%). It holds with no neighbouring region in training: a whole zone held out (5 zones) +1.9% / +2.7%, trained on the West and tested on the East and back +2.6% / +3.3%; tags +0.7% / −0.1% and +1.5% / −0.3%; terrain minus tags above zero in every draw.]

[TODO: R4: the hiker dominates whole-hike time (1,354 hikes in held-out regions; figure dropped for space, `fig_hike_time.png`): median error 13.6% (slope) → 13.0% (+ terrain); the hiker's first km brings it to 10.0%, and 9.6% with terrain.]

[TODO: R5, travel mode (plan claim 5): one curve times a per-mode multiplier predicts held-out runners as well as their own curve (×2.5; +0.1%), but not bikes and skis: their own curve is 3.8% better, and a constant ×3.2 is 15% too slow downhill and 35% too fast uphill.]

V. Discussion and Conclusion

[TODO: Why tags do not transfer (tested, decision #105): not because tag values mean different things. Within the same hike, harder sac_scale is slower in every region where it is used (−5% mountain_hiking vs hiking, 10 of 10 regions; −11% for harder ones, 8 of 8), and paved is faster (10 of 10). What does not carry over is whether a tag is there: sac_scale is on 1–96% of walked trail by region, and an untagged trail is faster than a tagged one in some regions and slower in others (−3.5 to +5.1% within a hike). So tags can only inform part of each region's trails, and how much differs by region; boosted trees

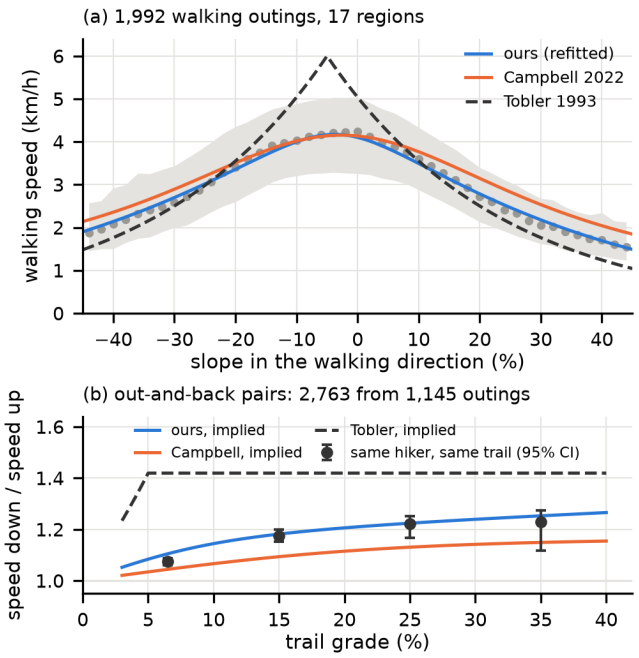


Fig. 2. (a) Walking speed vs slope in the walking direction, 17 regions: median per 2% bin (dots) and the middle 50% of segments (gray), our refitted curve, Campbell et al.'s published median curve [6] and Tobler's function [2]. (b) Speed down / speed up on out-and-back pairs (same hiker, same trail; grade bins 3–10, 10–20, 20–30, ≥30%), 95% intervals resampling outings, and the ratio each curve implies.

on tags also lose most in transfer (+1.5% → +0.4%), as if fitting region-specific tag combinations. Elevation-derived terrain exists on every trail, and its effect has the same sign in all 17 regions (−12%, steppiest vs smoothest fifth).]

[TODO: What terrain captures: uneven grade within the segment (rock steps, short pitches) that one average slope hides, plus difficulty beyond grade. It matters most going down and on the level, where footing sets the pace, and least on steep climbs, where effort does. STRIDE's micro-roughness, calibrated off-trail, over-penalizes trails.]

[TODO: Practical output: Eq. (1) runs anywhere a DEM exists (e.g. pace on all 3,369 km of White Mountains trail, 77% never walked in our data); travel-mode multipliers fail for bikes (ratio ∼4× on descents, ∼1.6× on climbs).]

[TODO: Limits: OSM uploaders are not a random sample of hikers; sparse traces read slow, a measurement effect: thinning dense walks to 30 s between points lowers measured speed by 11% (straight lines cut corners; short stops count as moving), and the curve without traces logged every >10 s peaks 3% higher (hiking functions depend on the sampling resolution [4]); distance along the mapped trail removes the corner cutting (walks thinned to 30 s keep 99.3–100.5% of their distance, vs 93.5–97.7% with straight lines) and would raise measured speeds by about 1–3%, but our numbers use straight lines; per-segment effects are a few percent because GPS noise and the hiker dominate; weather [22] slows hikers on hot days (−3% above 25°C) and with snow on the ground

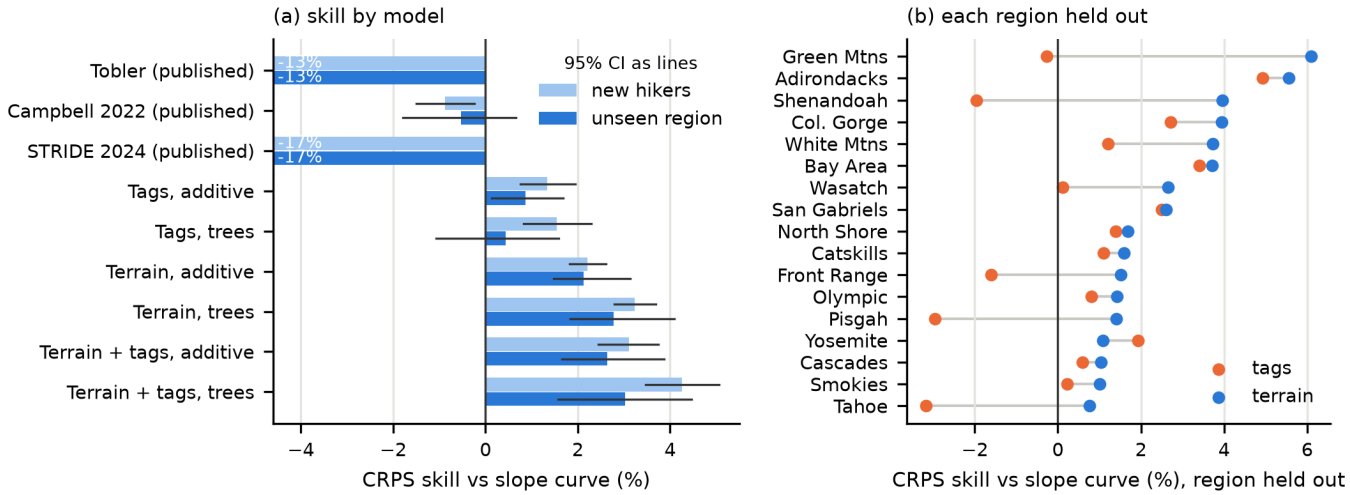


Fig. 3. CRPS skill vs the refitted slope curve. (a) By model, for new hikers in known regions (light) and in regions the model never saw (dark), with 95% intervals resampling whole regions. (b) Each region held out: terrain vs OSM tags (additive models); terrain beats tags in 16 of 17.

(−4%), as in Hong Kong [8], but neither weather nor date and time of day improve predictions in new regions (weather −0.0%; with terrain its gain drops to +1.9%), because the hiker's day-level pace already holds it.]

[TODO: Conclusion: three sentences. Data release (segments without user names) if the mentors agree.]

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